

[Assignment-1]
(Problem Set-1)
Full

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Course : CSE428
code

Section : 4

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Initial

[Lecture-1]

[Ans to the question no 1]

Image 1 (Chest X-Ray): Absorption.

X-rays pass through the body; dense tissues (bones) absorb radiation, while soft tissues allow it to pass, creating the image.

Image 2 (MRI Brain): Emission.

Magnetic Resonance Imaging (MRI) works by exciting protons which then emit radio frequency signals that are reconstructed into an image.

Image 3 (cat & Baby): Reflection.

This is a standard visible light photograph formed by light reflecting off the surfaces of the objects.

[Ans to the question no 2]

Let,

$$\text{Total pixels } (N) = 960\,000$$

$$\text{Aspect Ratio } (W:H) = 5:3$$

$$\text{Depth} = 8\text{-bit}$$

So,

$$\text{Total pixels} = 5x \times 3x = 15x^2$$

$$\Rightarrow 15x^2 = 960\,000$$

$$\Rightarrow x^2 = 64\,000$$

$$\therefore x \approx 252.98$$

$$\approx 253 \text{ [ceiling]}$$

So,

$$\text{columns} \approx 5 \times 253 = 1265$$

$$\text{Rows} \approx 3 \times 253 = 759$$

$$\text{Size of the image} = 960\,000 \times 8 = 7\,680\,000 \text{ bits}$$

$$= \frac{7\,680\,000}{8 \times 1024 \times 1024} \text{ MB}$$

$$= 0.9155 \text{ MB}$$

[Ans to the question no 3]

1D Signal: Function of a single variable.

Ex - Audio or EKG wave ($f(t)$)

2D Signal: Function of two variables, spatial coordinates

Ex - photograph ($f(x, y)$)

3D signal: Function of three variables, volumetric data or video.

Ex - CT scan or video ($f(x, y, t)$)

[Answer to the question no 4]

For,

1-based indexing: $f(1, 3) = 1$

0-based indexing: $f(1, 3) = 4$

[Answer to the question no 5]

Since, 4-bit image, so intensity values in the range $[0, 15]$

Tentative matrix:

$$\begin{bmatrix} 10 & 15 & 5 \\ 15 & 0 & 6 \\ 6 & 10 & 15 \end{bmatrix}$$

✓ center is black
(intensity ≈ 0)

[Lecture 2]

[Ans to the question no 1]

we know that,

$$s(x, y) = i(x, y) \times r(x, y)$$

$$\Rightarrow 400 = 800 \times r(x, y)$$

$$\therefore r(x, y) = 0.5$$

Reflectance $r(x, y) \in [0, 1]$ $\left| \begin{array}{l} 0 \text{ is total absorption,} \\ 1 \text{ " " reflection} \end{array} \right.$

Intensity $s(x, y) \in [0, L-1]$

[Ans to the question no 2]

The Weber Ratio indicates that human brightness discrimination is poor in very dark regions.

X-ray films historically appear negative (bones are white, background is black / transparent). Digitally inverting them (bones or displaying them as negative is often done to optimize the image for the human visual system's dynamic range, ensuring subtle density differences (contrast) fall within

the linear region of Weber's law where the eye is most sensitive.

[Ans to the question no 3]

The stages of a typical pipeline of digital image acquisition using a camera —

- 1) Illumination: Energy source lights the scene.
- 2) Scene Element: Reflects/ absorbs energy
- 3) Imaging System: Focuses energy (Lens) onto a sensor.
- 4) sensor: converts light energy into continuous electric signals.
- 5) Digitization: converts continuous analog signal to discrete digital data via sampling and quantization.

[Ans to the question no 4]

Dynamic Range: The ratio between the maximum and minimum measurable light intensities.

High Dynamic Range (HDR): Desirable because it allows an image

to capture details in both the brightest highlights and darkest shadows simultaneously without clipping.

Image (b) represents a higher dynamic range. It retains details in dark trees and bright snow, whereas Image (a) appears "washed out" with clipped highlights.

[Ans to the question no 5]

Sensors like CCD or CMOS contain array of photosites. When photons strike the silicon surface, they generate electron hole pair. This accumulated charge is converted into a voltage signal (analog) proportional to the light intensity.

[Ans to the question no 6]

Digitization

→ Sampling: Discretizing the spatial coordinates (x, y) .

Ex - deciding an image will be 1920×1080 pixels.

→ Quantization: Discretizing the amplitude/intensity values.

Ex - Assigning a voltage of $0.7V$ to integer value 178 in an 8-bit system.

Digitization is necessary because computers can only process discrete digital numbers, not continuous analog signals.

[Ans to the question no 7]

To avoid deterioration in image quality,
we can do -

a) False contouring: Caused by insufficient
Quantization (low bit-depth).
Large flat areas show visible bands
of color.

Solution: Increase bits per pixel.

b) Pixelation: Caused by insufficient
Sampling (low spatial resolution).
Individual pixels are visible.

Solution: Increase resolution (more pixels).

[Ans to the question no 8]

Given Image matrix (4x4):

$$\begin{bmatrix} 6 & 2 & 1 & 9 \\ 2 & 0 & 2 & 4 \\ 1 & 9 & 9 & 5 \\ 2 & 0 & 1 & 2 \end{bmatrix}$$

1) Upsampling to 8x8:

method: (Nearest Neighbor Interpolation)
so, 8x8 upsampled matrix:

$$\begin{bmatrix} 6 & 6 & 2 & 2 & 1 & 1 & 9 & 9 \\ 6 & 6 & 2 & 2 & 1 & 1 & 9 & 9 \\ 2 & 2 & 0 & 0 & 2 & 2 & 4 & 4 \\ 2 & 2 & 0 & 0 & 2 & 2 & 4 & 4 \\ 1 & 1 & 9 & 9 & 9 & 9 & 5 & 5 \\ 1 & 1 & 9 & 9 & 9 & 9 & 5 & 5 \\ 2 & 2 & 0 & 0 & 1 & 1 & 2 & 2 \\ 2 & 2 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

2) Downsampling to 2×2 :

Method: Local Averaging (Box Filtering)

Top-left Quadrant pixels: $\begin{bmatrix} 6 & 2 \\ 2 & 0 \end{bmatrix}$; $Avg = \frac{6+2+2+0}{4}$
 $= 2.5$
 $= 3 \text{ (ceiling)}$

Top-right Quadrant pixels: $\begin{bmatrix} 1 & 9 \\ 2 & 4 \end{bmatrix}$; $Avg = \frac{1+9+2+4}{4}$
 $= 4$

Bottom-left Quadrant pixels: $\begin{bmatrix} 1 & 8 \\ 2 & 0 \end{bmatrix}$; $Avg = \frac{1+8+2+0}{4}$
 $= 3$

Bottom-right Quadrant pixels: $\begin{bmatrix} 9 & 5 \\ 1 & 2 \end{bmatrix}$; $Avg = \frac{9+5+1+2}{4}$
 $= 4.25$
 $= 4 \text{ (ceiling)}$

So, 2×2 downsampled matrix: $\begin{bmatrix} 3 & 4 \\ 3 & 4 \end{bmatrix}$

[Ans to the question no 9]

Sensor → Retina
Lens → Optical Lens
Shutter → Iris & Pupil
Focusing Region → Fovea
Auto focus system /
Focus Ring → Ciliary Body

[Lecture 3: point processing (part 1)]

[Ans to the question no 1]

1.

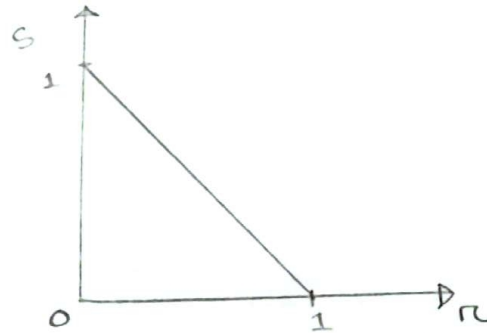
Relationship: $S = r$



Identity Transformation

2.

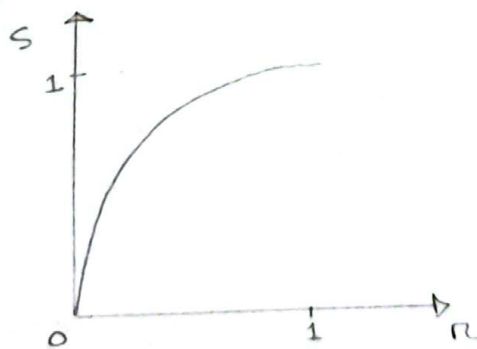
Relationship: $S = 1 - r$



Negative Transformation

3.

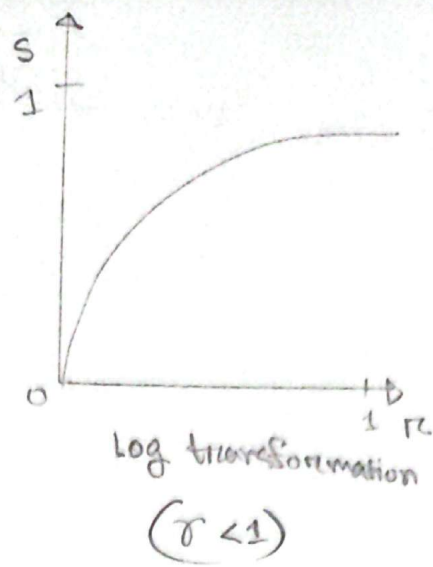
Relationship: $S = c \log(1+r)$
 $S = c r^r$
; $[r < 1]$



Log Transformation

$(r < 1)$

relationship:



[Ans to the question no 2]

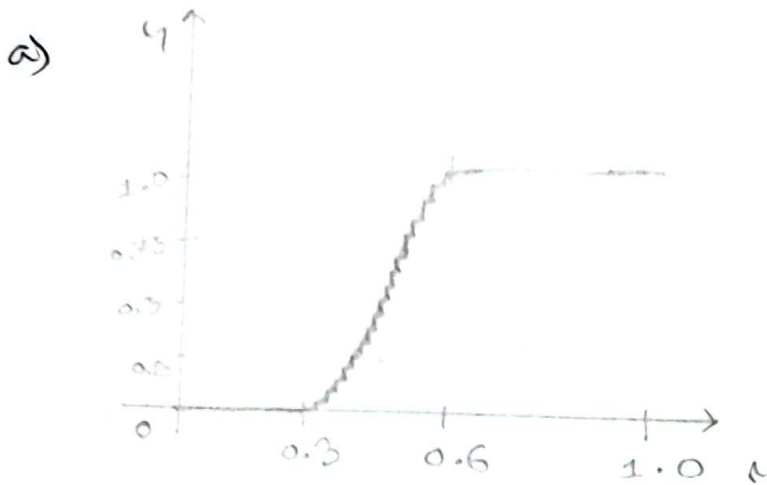
$$s = (2 * r + 6) / 255$$

- i) Brightness: Increase s . The $+6$ term adds a constant bias, shifting the histogram right.
- ii) Contrast : Increases. The factor 2 (slope > 1) spreads the intensity values apart.
- iii) Data Loss: Yes. Any pixel $[x]$ with $r > [x]$ will result in $s > 255$, causing clipping/saturation.

Here,

$$\begin{aligned} s &= (2 * r + 6) / 255 \\ \Rightarrow 1 &= (2 * r + 6) / 255 \\ \therefore r &= 124.5 \approx 124 = x \end{aligned}$$

[Ans to the question no 3]



CDF

b)

Input range $[0.3, 0.6]$

Target " $[0, 1]$

we know that,

$$s = \frac{\pi - \pi_{\min}}{\pi_{\max} - \pi_{\min}} = \frac{\pi - 0.3}{0.6 - 0.3} = \frac{\pi - 0.3}{0.3} = \frac{1}{0.3}(\pi - 0.3)$$
$$\approx 3.33(\pi - 0.3)$$

$\uparrow \qquad \qquad \uparrow$
 $c_1 \qquad \qquad c_2$

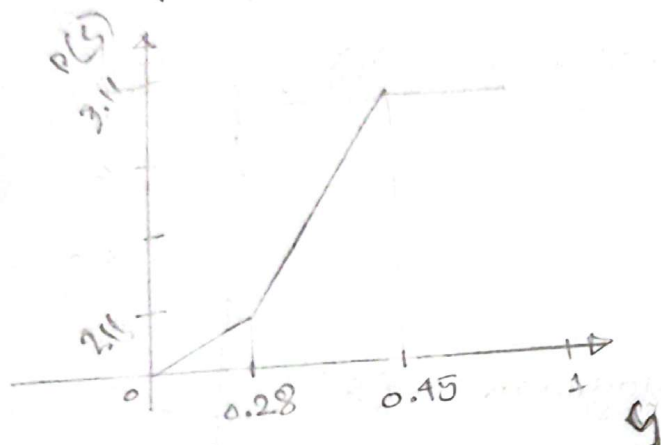
$$\text{So, } c_1 = 3.33$$

$$c_2 = 0.3$$

$$c) S = \log(1 + 3I)$$

$$S_{\text{start}} = \log(1 + 3 \times 0.3) \approx 0.28$$

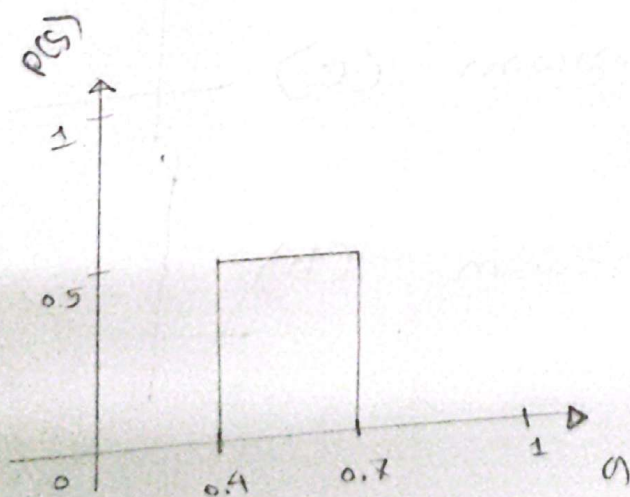
$$S_{\text{end}} = \log(1 + 3 \times 0.6) \approx \cancel{0.6445} 0.45$$



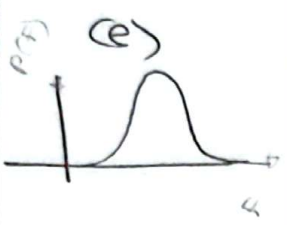
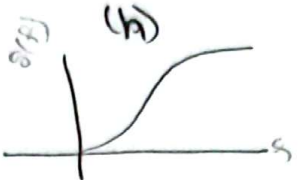
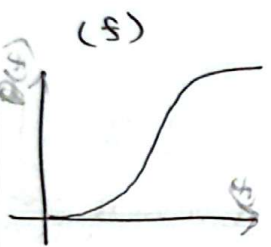
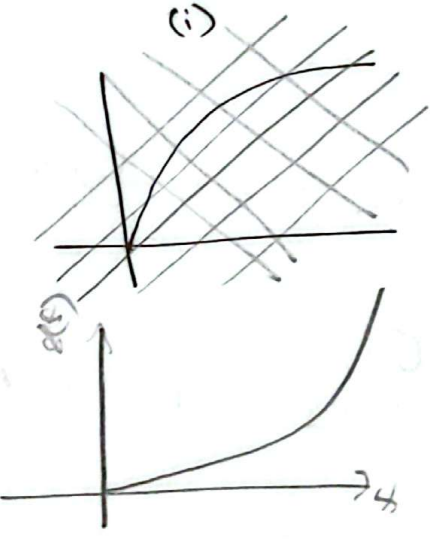
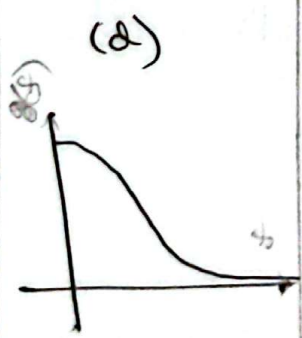
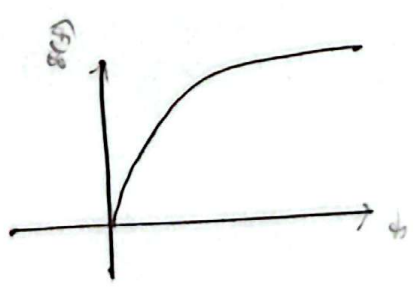
$$d) S = 1 - I$$

$$S_{\text{start}} = 1 - 0.3 = 0.7$$

$$S_{\text{end}} = 1 - 0.6 = 0.4$$

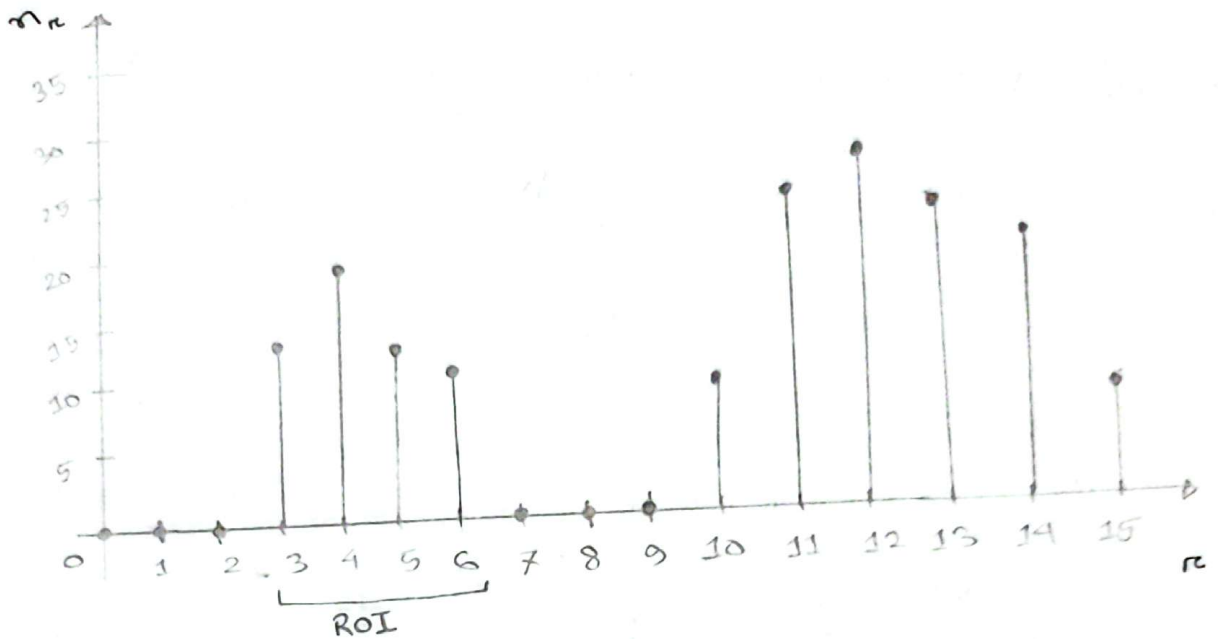


[Ans to the Question no 4]

Image	Histogram	Transformation	Explanation
(a)			The original Image has balanced brightness levels, corresponding to the bell shaped histogram centered. The mildly curved transformation provides general contrast enhancement without over stretching any particular range.
(b)			The brighter version has most pixels at higher intensity values, matching the increasing histogram skewed towards brighter tones. Transformation (g) is a concave-up curve. It keeps low values low and suppresses bright values until the very end of the range effectively darkening the image.
(c)			The darker version has most pixels at lower intensity values, corresponding to the decreasing histogram skewed towards darker tones. Transformation (i) is a concave-down curve. It maps low input values to much higher output values, boosting brightness.

[Ans to the question no 5]

a)



Region of Interest (ROI) : $n \in [10, 15]$ $[3, 6]$
(Darker ~~values~~ pixels)

b) Stretch: map $[3, 6] \rightarrow [0, 15]$

$$\text{slope, } c_1 = \frac{15-0}{15-10} = \frac{15}{5} = 3$$

Start point implies $0 = 3(10 - c_2)$

$$\Rightarrow 10 - c_2 = 0$$

$$\therefore c_2 = 10$$

$$\therefore T(n) = 3 \times (n - 10)$$

b) stretch: $\text{map}[3, 6] \longrightarrow [0, 15]$

$\uparrow \quad \uparrow \quad \quad \uparrow \quad \uparrow$
 $r_{\min} \quad r_{\max} \quad \quad s_{\min} \quad s_{\max}$

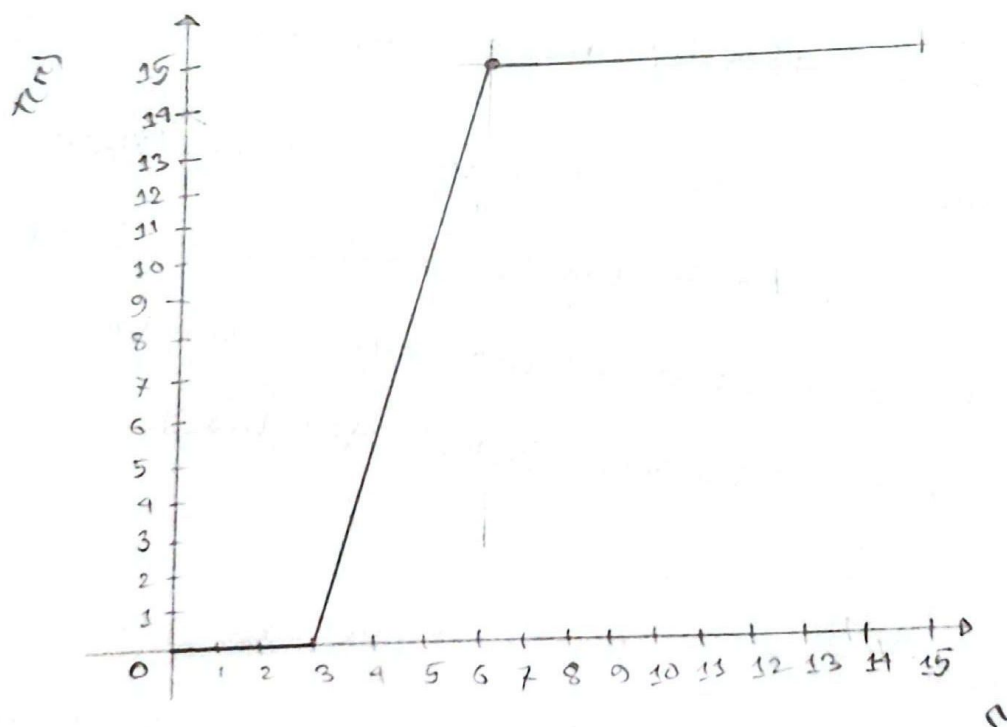
$$T(r) = c_1 \times (r - c_2)$$

$$\Rightarrow 0 = c_1 \times (3 - c_2)$$

$$c_2 = r_{\min} = 3$$

$$c_1 = \frac{s_{\max} - s_{\min}}{r_{\max} - r_{\min}} = \frac{15 - 0}{6 - 3} = \frac{15}{3} = 5$$

$$T(r) = 5(r - 3)$$



c) $\gamma > 1$

In the original image, the numbers "428" are dark grey. In the transformed image shown in part (c), the numbers "428" have become solid black (intensity ≈ 0).

Effect of Gamma.

$\gamma < 1$: Expands dark values, making the image brighter and "washed out". This would make the "428" lighter grey.

$\gamma = 1$: Linear Identity transform (no change)

$\gamma > 1$: Compresses low intensity values towards 0, making dark pixels darker. It increases contrast in the bright areas while crushing shadow detail to black.

Since the dark "428" pixels have been pushed to black (darkened), the transform must have used $\gamma > 1$.

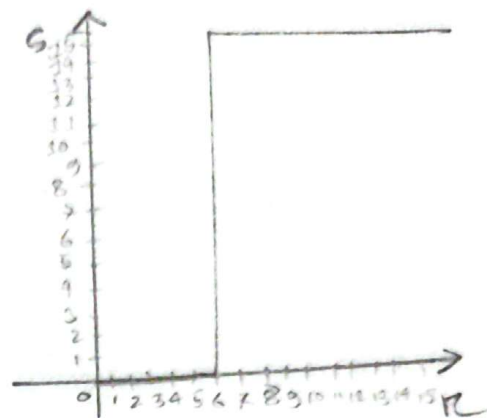
d) This is a thresholding function.

The "428" is black (intensity 0) and the background is white (intensity 15)

Intensity map:

$$s = 0 \text{ for } 0 \leq r \leq 6$$

$$s = 15 \text{ for } 7 \leq r \leq 15$$



e) Piecewise Linear Transformation:

a)

Given,

$$(r_1, s_1) = (9, 2)$$

$$(r_2, s_2) = (13, 14)$$

Region-1

$$(0, 0) \text{ to } (9, 2) \rightarrow m_1 = \frac{2-0}{9-0} = \frac{2}{9} \approx 0.22$$

$$[0 \rightarrow r_1]$$

Region-2

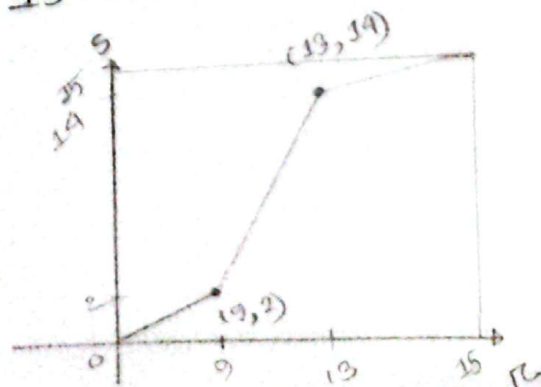
$$(9, 2) \text{ to } (13, 14) \rightarrow m_2 = \frac{14-2}{13-9} = \frac{12}{4} = 3$$

$$[r_1 \rightarrow r_2]$$

Region-3

$$(13, 14) \text{ to } (15, 15) \rightarrow m_3 = \frac{15-14}{15-13} = \frac{1}{2} = 0.5$$

$$[r_2 \rightarrow 15]$$



(b)

Region 1 ($0 \leq \pi \leq 9$): passes through origin (0,0)

$$s_0, c = 0$$

$$s = 0.22\pi$$

Region 2 ($9 < \pi \leq 13$):

$$s - s_1 = m_2 (\pi - \pi_1)$$

$$\Rightarrow s - 2 = 3 (\pi - 9)$$

$$\Rightarrow s - 2 = 3\pi - 27$$

$$\Rightarrow s = 3\pi - 25$$

$$\therefore s = 3\pi - 25$$

Region 3 ($13 < \pi \leq 15$):

$$s - s_2 = m_3 (\pi - \pi_2)$$

$$\Rightarrow s - 14 = 0.5 (\pi - 13)$$

$$\Rightarrow s = 0.5\pi - 6.5 + 14$$

$$\therefore s = 0.5\pi + 7.5$$

②

Input (n)	Equation Used	calculation	Output (S) (After rounding)
0	$S = 0.22n$	0	0
1	"	0.22	0
2	"	0.44	0
3	"	0.66	1
4	"	0.88	1
5	"	1.11	1
6	"	1.33	1
7	"	1.55	2
8	"	1.77	2
9	Given	Exact	2
10	$S = 3n - 25$	5	5
11	"	8	8
12	"	11	11
13	Given	Exact	14
14	$S = 0.5n + 7.5$	14.5	15
15	"	15	15

[Lecture 4: Histogram Equalization]

[Ans to the question no 1]

a)

1	2	0	0
1	0	1	3
0	1	0	1
2	0	1	1

r_k	$h(r_k)$	$S_k(r_k)$
0	6	$6/16 = 0.375$
1	7	$7/16 = 0.4375$
2	2	$2/16 = 0.125$
3	1	$1/16 = 0.0625$

b)

$H(r_k)$	$F(r_k)$
6	0.375
13	0.8125
15	0.9375
16	1.0

c)

$$S_k = \text{round}((L-1) \times F(r_k))$$

r_k	S_k
0	$\text{round}(3 \times 0.375) = \text{round}(1.125) = 1$
1	$\text{round}(3 \times 0.8125) = 2$
2	$\text{round}(3 \times 0.9375) = 3$
3	$\text{round}(3 \times 1.0) = 3$

So,

Mapping: $0 \rightarrow 1$
 $1 \rightarrow 2$
 $2 \rightarrow 3$
 $3 \rightarrow 3$

d)

Original Image:

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 0 & 1 & 3 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 1 & 1 \end{bmatrix}$$

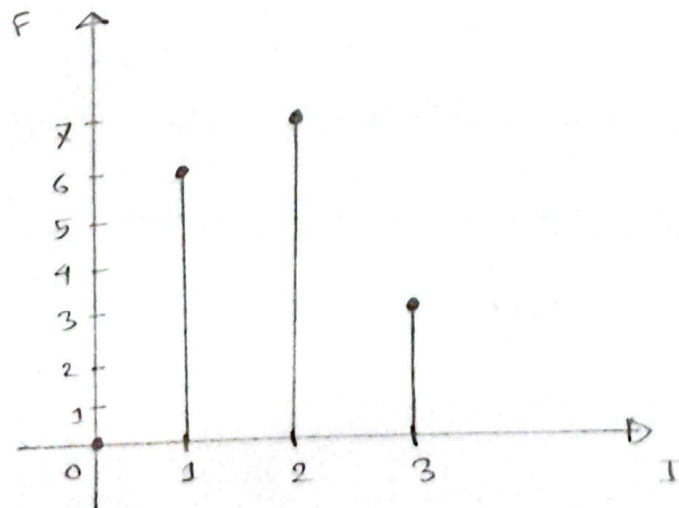


Equalized Image:

$$\begin{bmatrix} 2 & 3 & 1 & 1 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 2 \\ 3 & 1 & 2 & 2 \end{bmatrix}$$

e)

Intensity	Frequency
0	0
1	6
2	7
3	3



[Ans to the question no 2]

⑥

r	$n(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$	$F(r_k)$ [CDF]	HE Pixel
0	6	$6/196 \approx 0.03$	6	0.03	$\text{round}(x \times 0.03)$
1	0	0	6	0.03	$0.21 \approx 0 \rightarrow 0.21 \neq 0$
2	41	$41/196 \approx 0.21$	47	0.24	$1.68 \rightarrow 2$
3	66	$66/196 \approx 0.34$	113	0.58	$4.06 \rightarrow 4$
4	42	$42/196 \approx 0.21$	155	0.79	$5.93 \rightarrow 6$
5	23	$23/196 \approx 0.12$	178	0.91	$6.67 \rightarrow 6$
6	$x \rightarrow 11$	$11/196 \approx 0.06$	189	0.96	$6.72 \rightarrow 7$
7	7	$7/196 \approx 0.04$	196	1.00	$7.00 \rightarrow 7$

a) Total pixels = 14×14
 $= 196$

Sum of total counts = $6 + 0 + 41 + 66 + 42 + 23 + 7 + x$
 $= 185 + x$

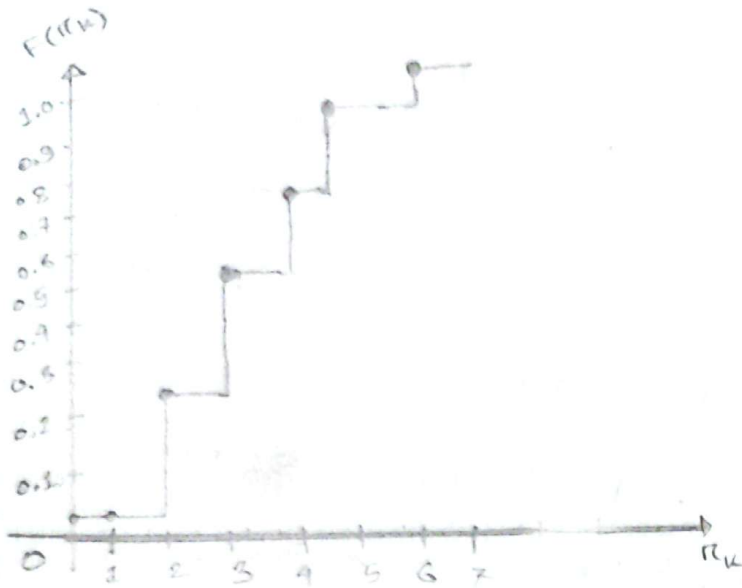
So, $185 + x = 196$

$\therefore x = 11$

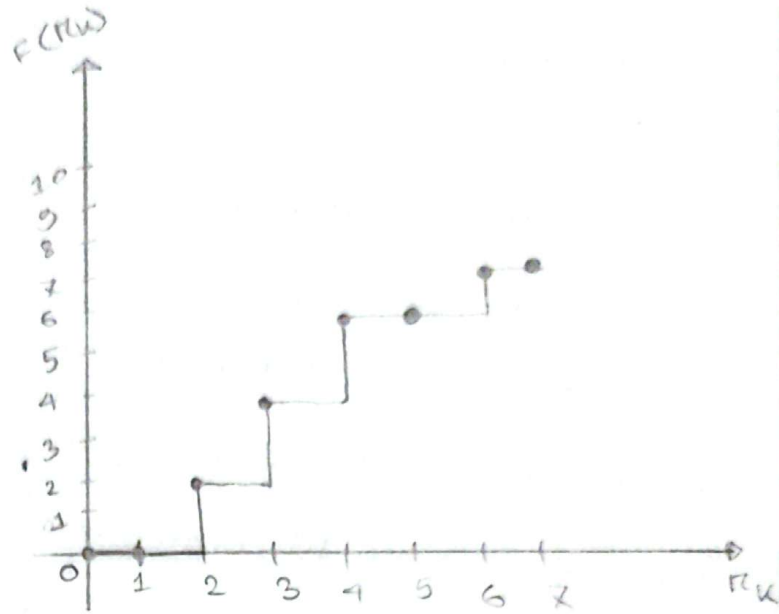
(Ans)

c)

Original CDF



Equalized CDF



The histogram equalization forces the CDF to become straight diagonal line (linear). It spreads the clustered intensity values (originally bunched around 2,3,4) out across the full available range (mapping them to 2,4,6).

In information theory, Entropy is a measure of randomness or information content. A perfectly uniform distribution (where every intensity level is equally probable) has maximum Entropy.

By equalizing the histogram, we are approximating a uniform distribution. Therefore, Histogram Equalization tends to increase the entropy of the image, visible detail and contrast as information.

[Ans to the question no 3]

a) CLAHE was used here.

Global HE would process the entire image based on a single histogram. Since the original image is dominated by dark pixels, AHE would stretch these dark values across the entire intensity range. AHE computes histograms for local regions. While this improves local contrast, it has a major tendency to over-amplify noise in relatively uniform regions, such as dark background in the image.

The resulting image here appears remarkably clean with sharp details and no visible noise explosion in the dark areas. CLAHE mitigates this by clipping the histogram in each tile to limit excessive contrast amplification, resulting in a balanced, noise-reduced enhancement that matches the provided equalized image.

b)

AHE: The image would likely to lose its artistic, moody look. The large amount of dark background would dominate the global histogram, causing the algorithm to over-stretch the shadows. The dark background would become a noisy.

AHE: The dog's face would look similar to the current result but the smooth, dark background areas would likely suffer from severe noise amplification.

c) CLAHE improves over AHE by preventing the over-amplification of noise.

In a local tile that is mostly uniform the histogram is a high spike. Equalizing this spike creates a mapping function with a very steep slope, which amplifies any tiny variations (noise) in that tile into visible artifacts.

CLAHE sets a clip limit on the height of the histogram bins. Before calculating the CDF, any histogram bin that goes above this limit is clipped. The clipped pixels are then redistributed equal among the other bins. The process limits the

slope of the transformation function, thereby restricting the maximum amount of contrast stretching and effectively suppressing noise generation.

— x —